

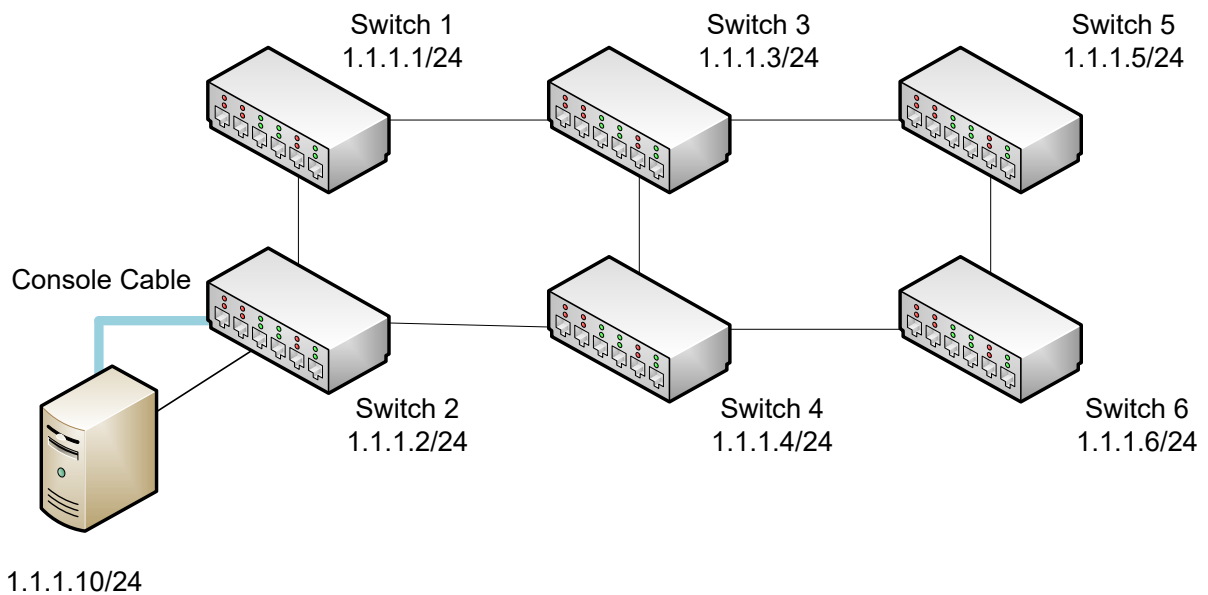
Lab 9

STP Operation

Objective:

The purpose of this lab is for the student to understand basic STP functionality. Students will also utilize IOS commands to explore STP functionality and features.

Diagram:



Components:

6 – Switches

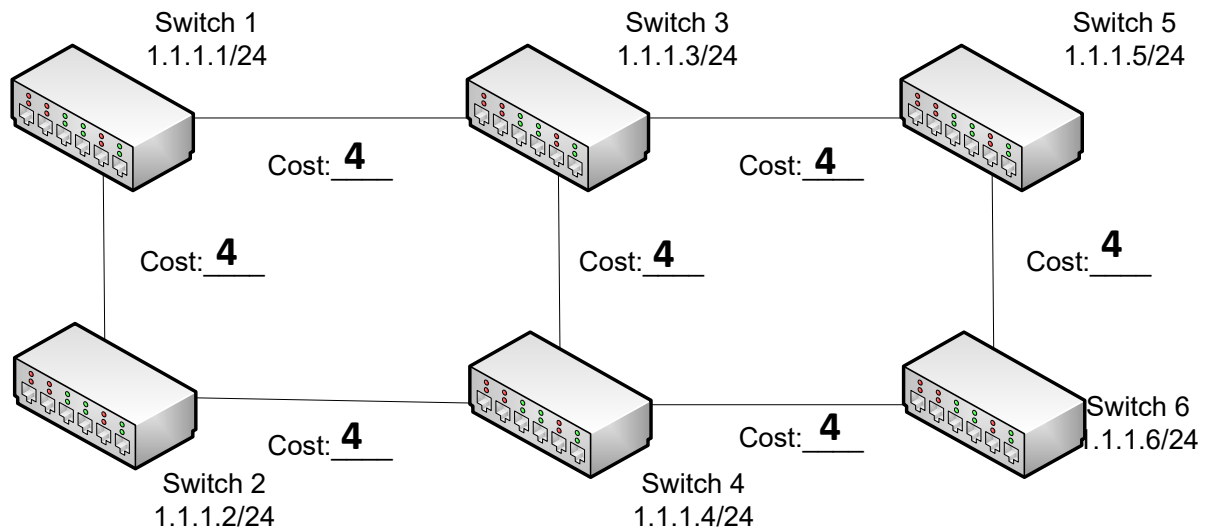
Procedure:

1. Connect the PC and the switches together and put their static IP addresses on them. Verify connectivity by pinging each switch from the PC.
2. Find the MAC address of each switch
3. List the MAC addresses of each switch below

- a. Switch 1 MAC address
- b. Switch 2 MAC address
- c. Switch 3 MAC address
- d. Switch 4 MAC address
- e. Switch 5 MAC address
- f. Switch 6 MAC address

```
C:\>arp -a
Internet Address      Physical Address
1.1.1.1               0030.f246.457b
1.1.1.2               0002.1647.1977
1.1.1.3               00e0.f90a.6c75
1.1.1.4               0040.0bce.3967
1.1.1.5               000c.cf7c.78b5
1.1.1.6               0060.70cc.98c0
```

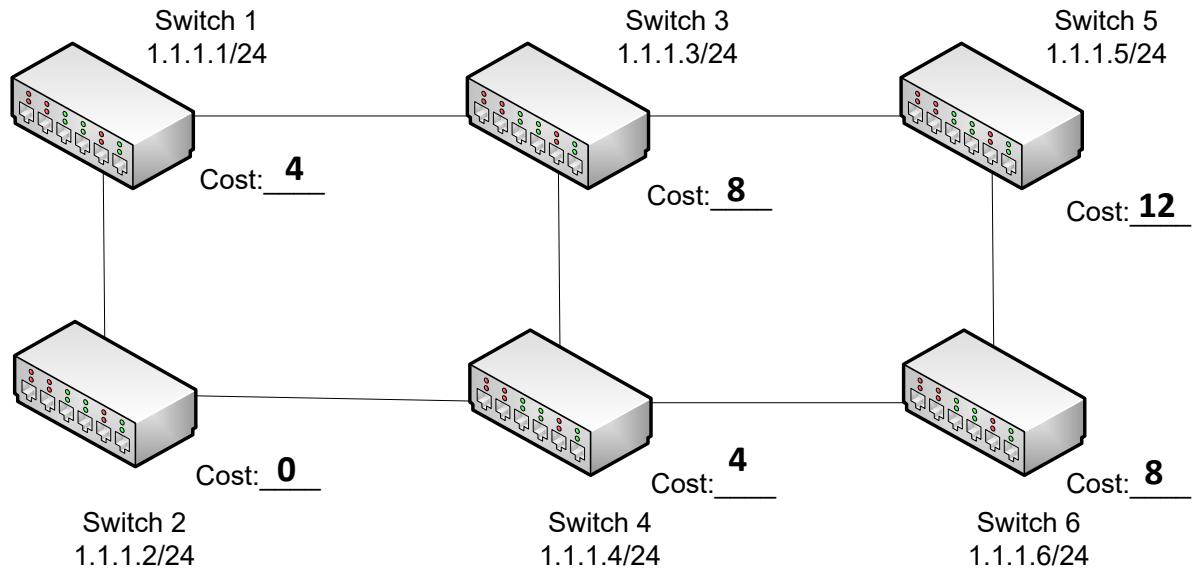
4. Which switch will be the root switch? Switch 2
 - a. Why? It has the lowest MAC address, and the default BPDU priorities have not been changed.
5. Which switch(s) would be the best root switch? The two in the middle – 3 or 4.
 - a. Why? They would require the least amount of hops or jumps between the devices.
6. Label the path costs to the root switch on the diagram below.



7. List the root port of each switch below

- a. Switch 1 root port – Fa0/24
- b. Switch 2 root port – This is the root switch
- c. Switch 3 root port – Fa0/23
- d. Switch 4 root port – Fa0/23
- e. Switch 5 root port – Fa0/24
- f. Switch 6 root port – Fa0/23

8. Label the total path costs to the root switch for each switch the diagram below.



9. View STP information on switch 1.

- What command did you enter? show spanning-tree
- Is this switch the root? no
- How can you tell? It has a port designated for the root
- What is the switches root port? Fa0/24

10. View STP information on switch 2.

- What command did you enter? show spanning-tree
- Is this switch the root? yes
- How can you tell? It says "this bridge is the root"
- What is the switches root port? N/A

11. View STP information on switch 3.

- What command did you enter? show spanning-tree
- Is this switch the root? no

- c. How can you tell? It has a root port
- d. What is the switches root port? Fa0/23

12.View STP information on switch 4.

- a. What command did you enter? show spanning-tree
- b. Is this switch the root? no
- c. How can you tell? It has a root port
- d. What is the switches root port? Fa0/23

13.View STP information on switch 5.

- a. What command did you enter? show spanning-tree
- b. Is this switch the root? no
- c. How can you tell? It has a root port
- d. What is the switches root port? Fa0/24

14.View STP information on switch 6.

- a. What command did you enter? show spanning-tree
- b. Is this switch the root? no
- c. How can you tell? It has a root port
- d. What is the switches root port? Fa0/23

15.Make switch 3 the root switch.

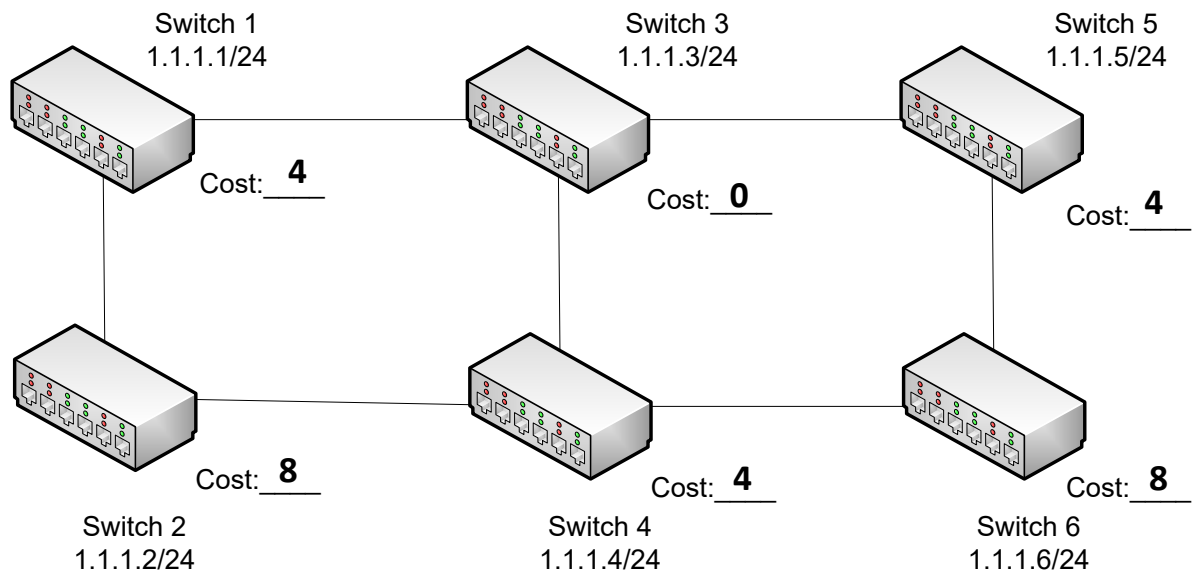
- a. What command did you enter? spanning-tree vlan 1 priority 0

16. List the new root port of each switch below

- a. Switch 1 root port – Fa0/23
- b. Switch 2 root port – Fa0/24
- c. Switch 3 root port – This is the root switch

- d. Switch 4 root port – Fa0/24
- e. Switch 5 root port – Fa0/23
- f. Switch 6 root port – Fa0/24

17. Label the total path costs to the new root switch for each switch the diagram below.



18. Enable telnet access to each switch from the PC. Configure the enable password of **cisco** and a telnet password of **access**

- a. What commands did you enter?
- b. Config terminal
- c. Enable password cisco
- d. Line vty 0 4
- e. Password access
- f. Login

19. Why would we want to enable telnet access to the switches?

Ease of access – it can be inconvenient to have to physically plug into switches

20.What remote access protocol is more secure than telnet? SSH

21.Unplug the network cable of PC from switch 1 and plug it back in

22.What color is the switchport light? orange

23.View the STP port state of the port the PC is plugged into.

a. What command did you enter? show spanning-tree

24.What state is the port in? LSN

25.Keep executing the STP command.

a. Do you observe the different port states? Yes – LRN, FWD

26.Keep executing the STP command until the port state is FWD.

a. What port state is this? Forwarding

b. How long with the port stay in this state? Until the connection disconnects, I suppose.

c. Can a device utilize the network if its port is orange? No, orange indicates that it's in block mode.

d. How long does it take the switchport to go from orange to green? 50 seconds.

27. Draw the lab diagram below (PC's and hub connections) Label the IP addresses, and subnet masks of each device. Feel free to use packet tracer or Visio, but do not copy and paste the diagram at the beginning of the lab. Screen shot your diagram and paste it below. Label the root switch, root ports, and costs.

